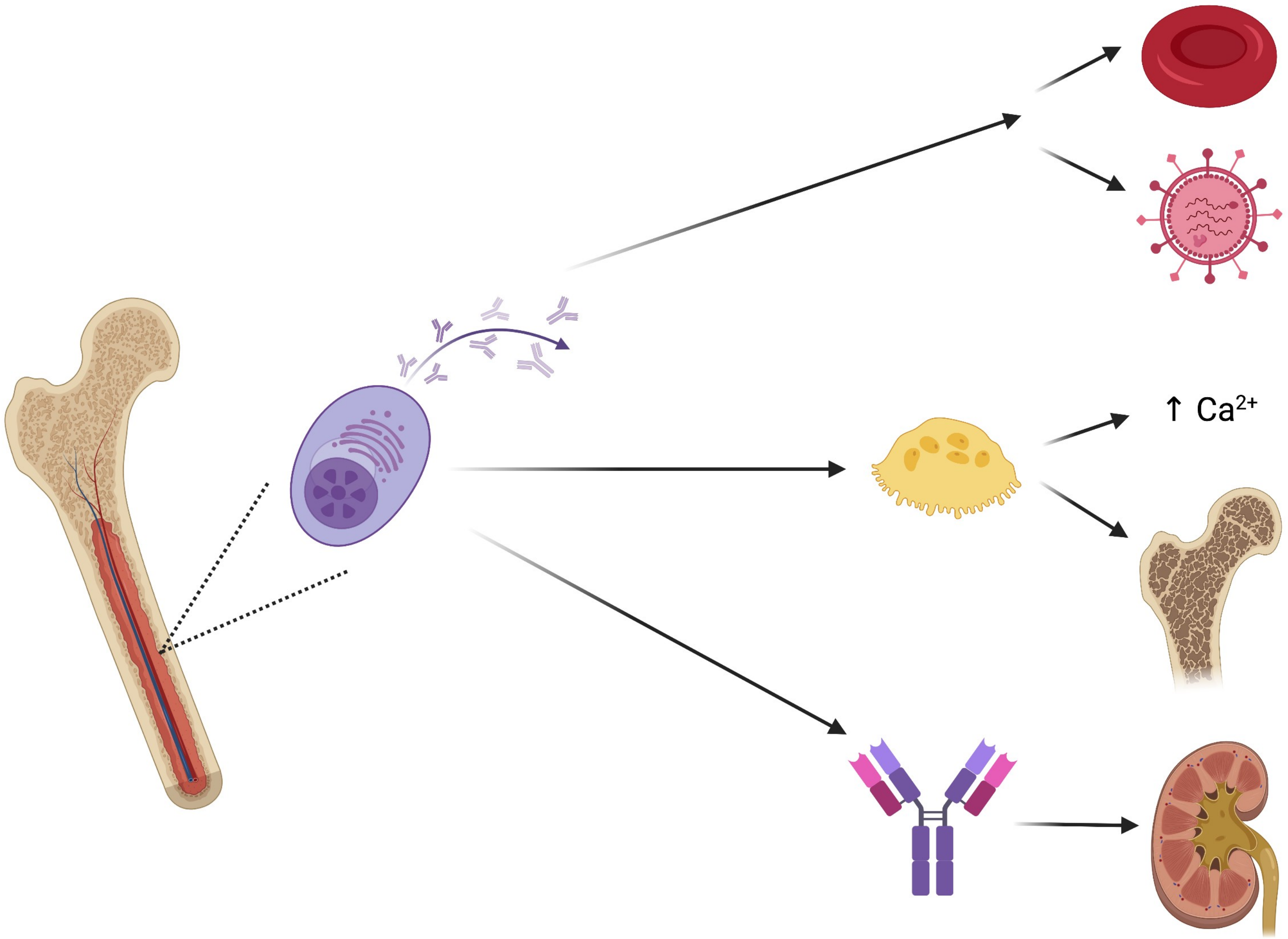
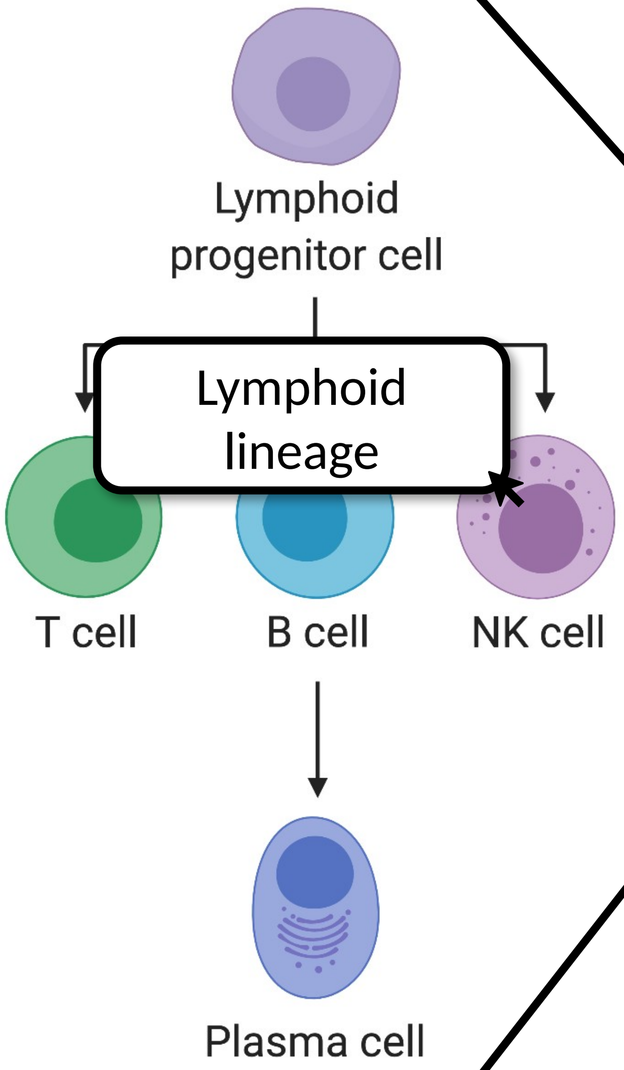
- Pathogenesis
- Testing
- Differential
- Prognosis
- Treatment
- Cases

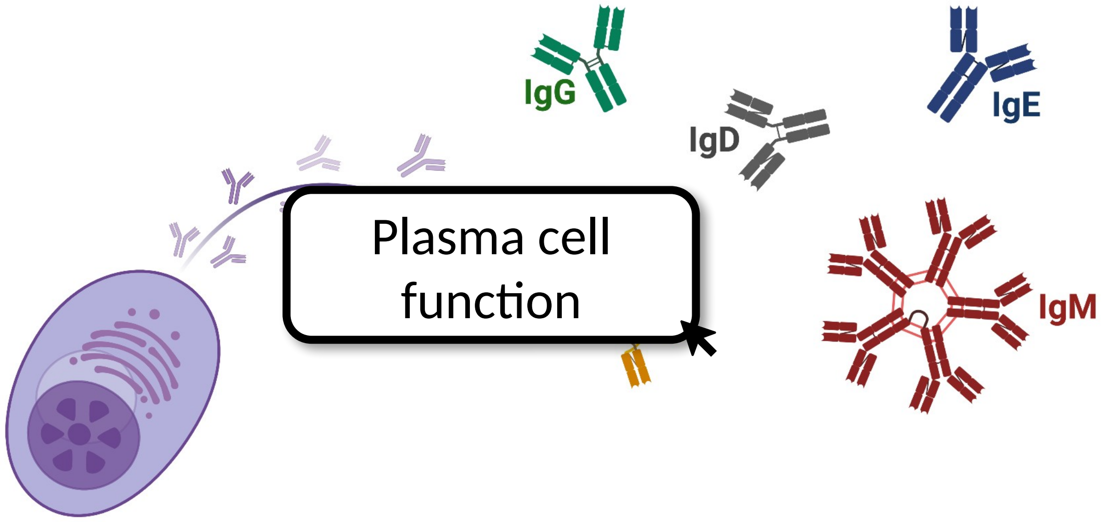
Plasma Cell Dyscrasias

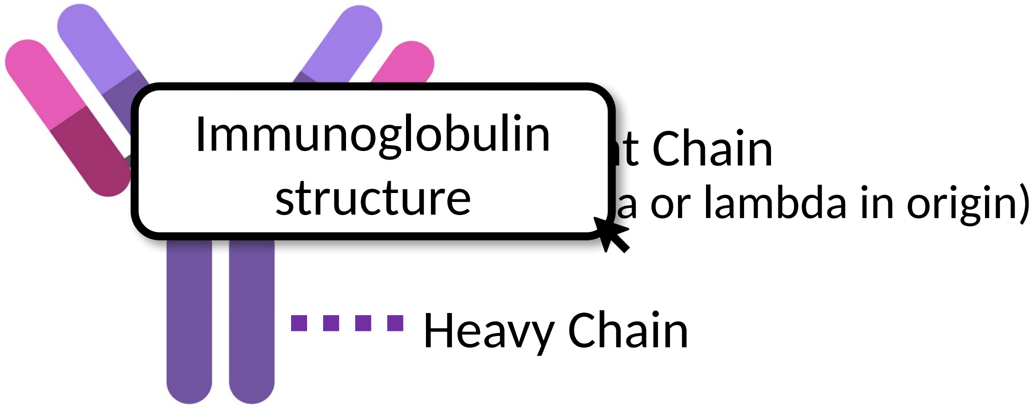


- Pathogenesis
  - Basics
  - Pathology
- Testing
- Differential
- Prognosis
- Treatment
- Cases

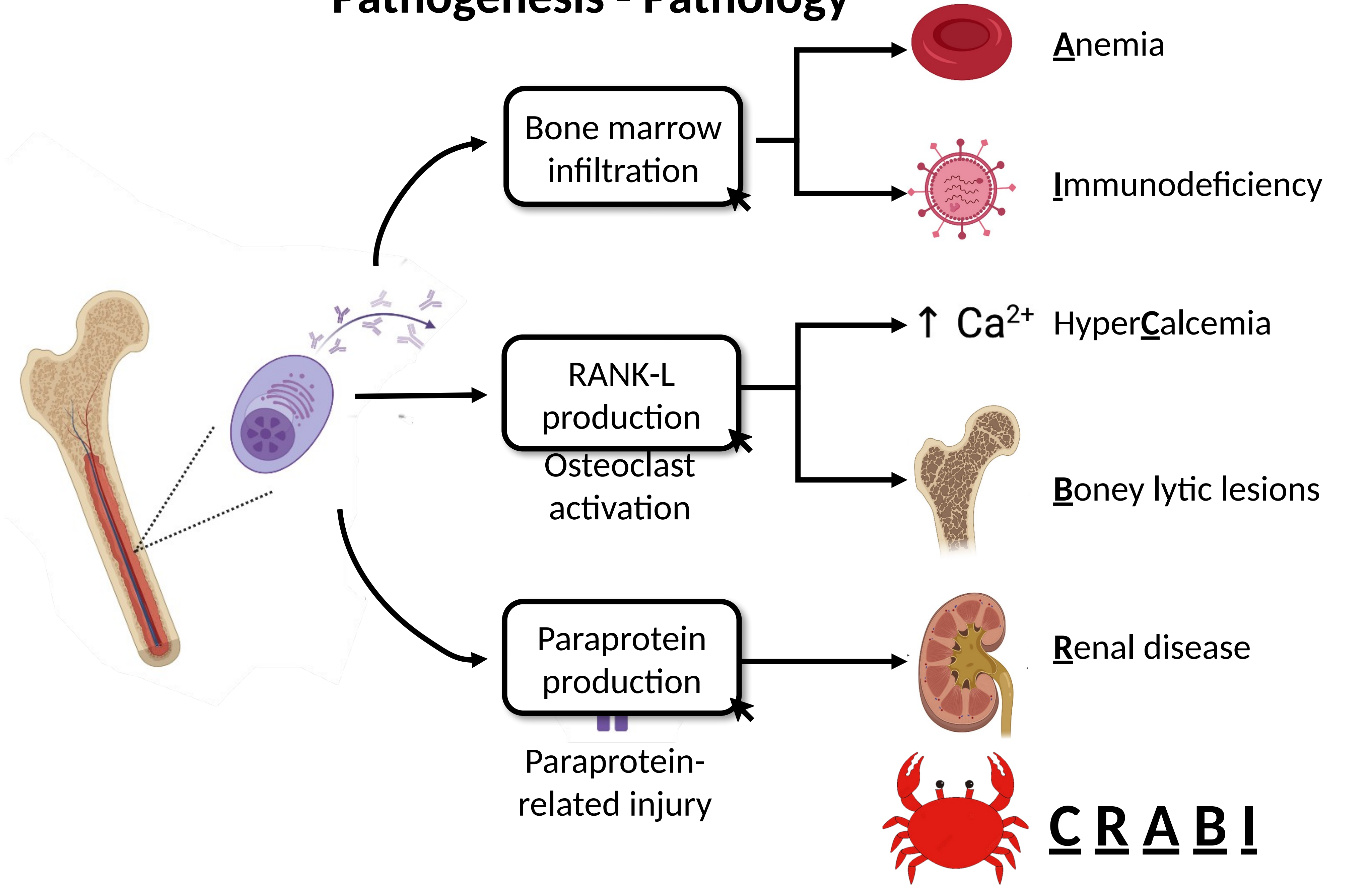
Pathogenesis - Basics







Pathogenesis - Pathology



Pathogenesis

Testing

- SPEP
- UPEP
- FLC

Differential

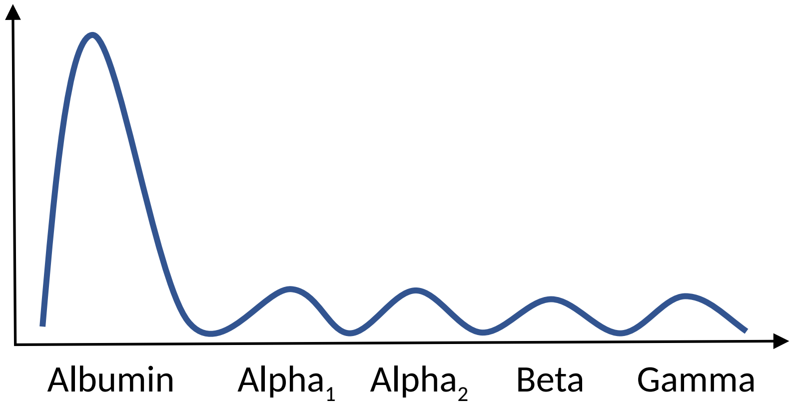
Prognosis

Treatment

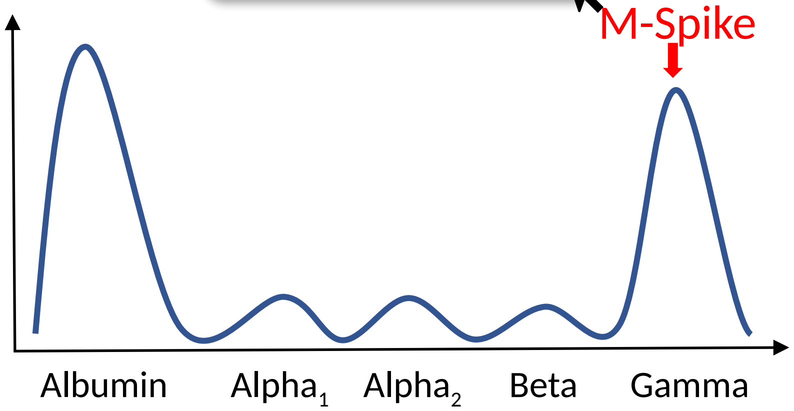
Cases

Testing – Serum Protein Electrophoresis

Normal SPEP



Abnormal SPEP



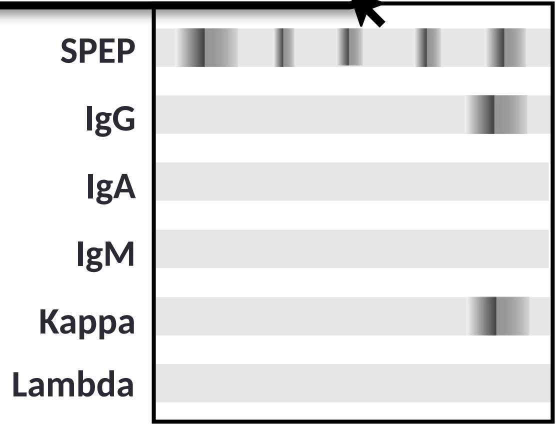
Example SPEP

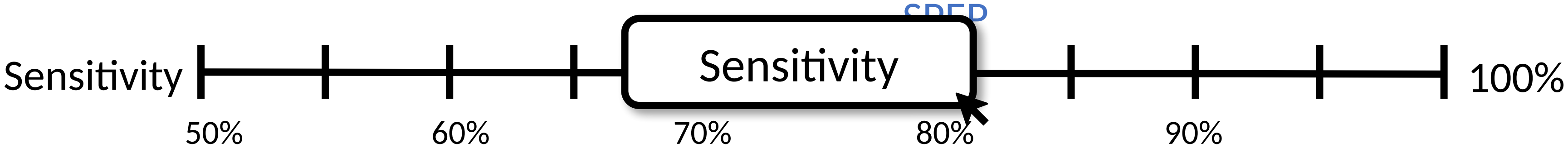
	Result (g/dL)	Reference Range (g/dL)
Albumin	4.2	3.6 -5.4
Alpha$_1$ globulin	0.3	0.1-0.3
Alpha$_2$ globulin	0.3	0.5-0.3
Beta globulin	1	0.5-1
Gamma globulin	4.5 (H)	0.5-1.4

"The monoclonal protein measures 3.2 g/dL"

Example Immunofixation

"There is an IgG Kappa restricted M-protein "





Pathogenesis

Testing

SPEP

UPEP

FLC

Differential

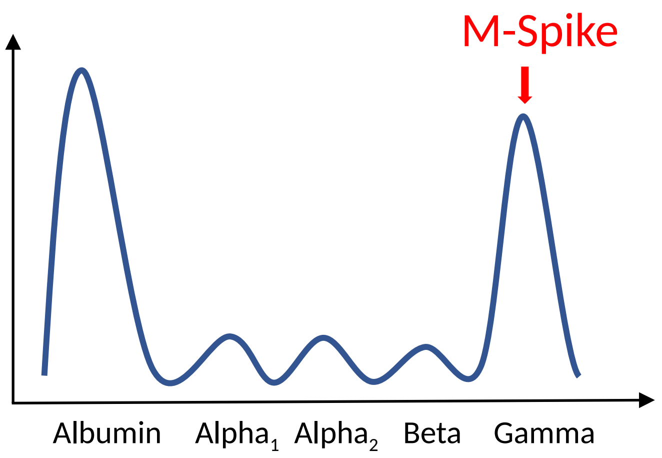
Prognosis

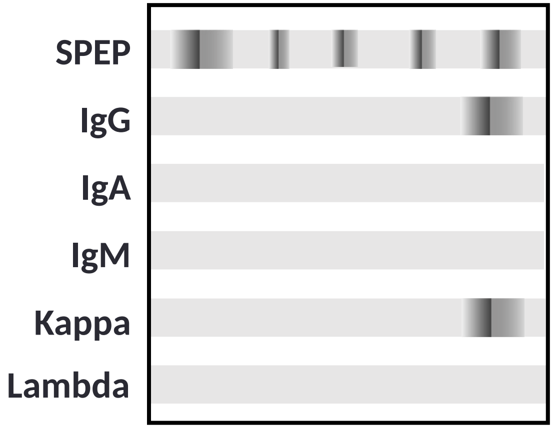
Treatment

Cases

Testing – Urine Protein Electrophoresis

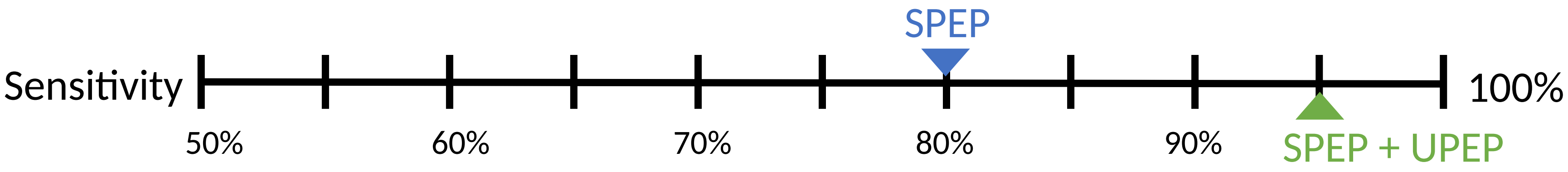
Obtained from 24 h urine sample



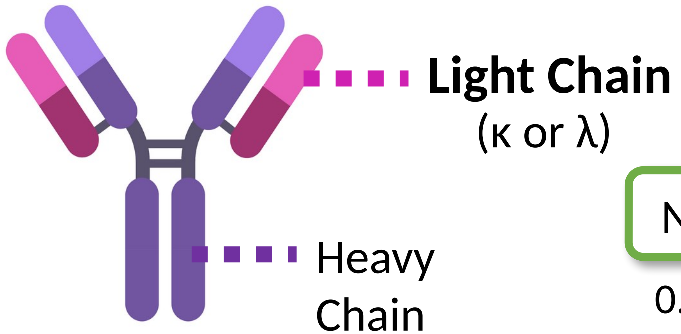


Why obtain a UPEP?

UPEP improves sensitivity of SPEP from 80% to ~ 95%



Testing – Free Light Chains

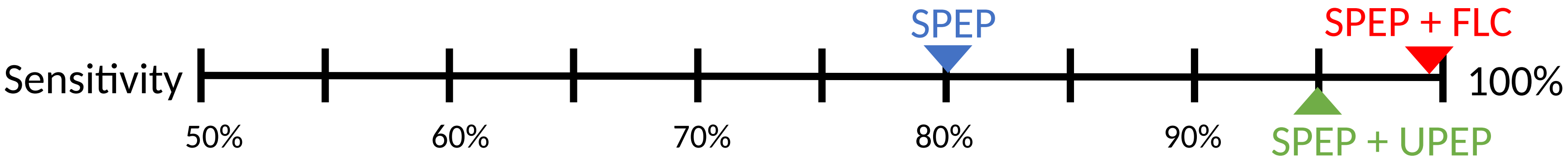


Measures $\frac{\kappa \uparrow\uparrow}{\lambda \uparrow\uparrow}$ *Normally produced in ~equal amounts* / *↓ clearance of light-chains*

Normal	κ disease	λ disease	ESRD	MM
0.26-1.65	>1.65	<0.26	0.33-3.0	>100 or <0.01

Why obtain FLC?

1. Up to 16% have “light chain” only disease
2. FLC (in conjunction with SPEP) ↑ sensitivity and can replace UPEP



Pathogenesis

Testing

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Differential Diagnosis

	MGUS	Smoldering Myeloma	Multiple Myeloma
M protein (serum)	<3 g/dL	≥3 g/dL	≥3 g/dL*
M protein (urine)	<500 mg/24 h	≥500 mg/24 h	≥500 mg/24 h
% clonal plasma cells (bone marrow)	<10%	10-60%	≥10% (≥60% is diagnostic)
End organ damage	None	None	CRABI
Other MM defining events	None	None	- Extramedullary plasmacytoma - FLC > 100mg/dl and FLC ratio >100 or < 0.01 - >1 focal lesion on MRI
	1% ... prog ... MM	... of ... n to MM in first 5 y	*Oligo-secretory and non-secretory MM subtypes will not meet M-protein criteria

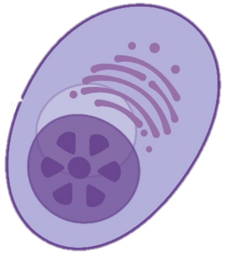
Why should we care?

- Pathogenesis
- Testing
- Differential
- **Prognosis**
- Treatment
- Cases

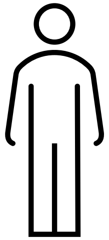
Prognosis

R-ISS Criteria	**Stage I**	**Stage II**	**Stage III**
β2- microglobulin	<3.5 mg/L	Not meeting criteria for R-ISS stage I or III	>5.5 mg/L
	+		+
LDH	Normal		↑ LDH
	+		±
iFISH	Standard risk		High risk del(17p), t(4;14), t(14;16)
	+		
Albumin	≥3.5 g/dL		
Median Survival	82% survival at 5 years	83 months	43 months

Other prognostic factors?



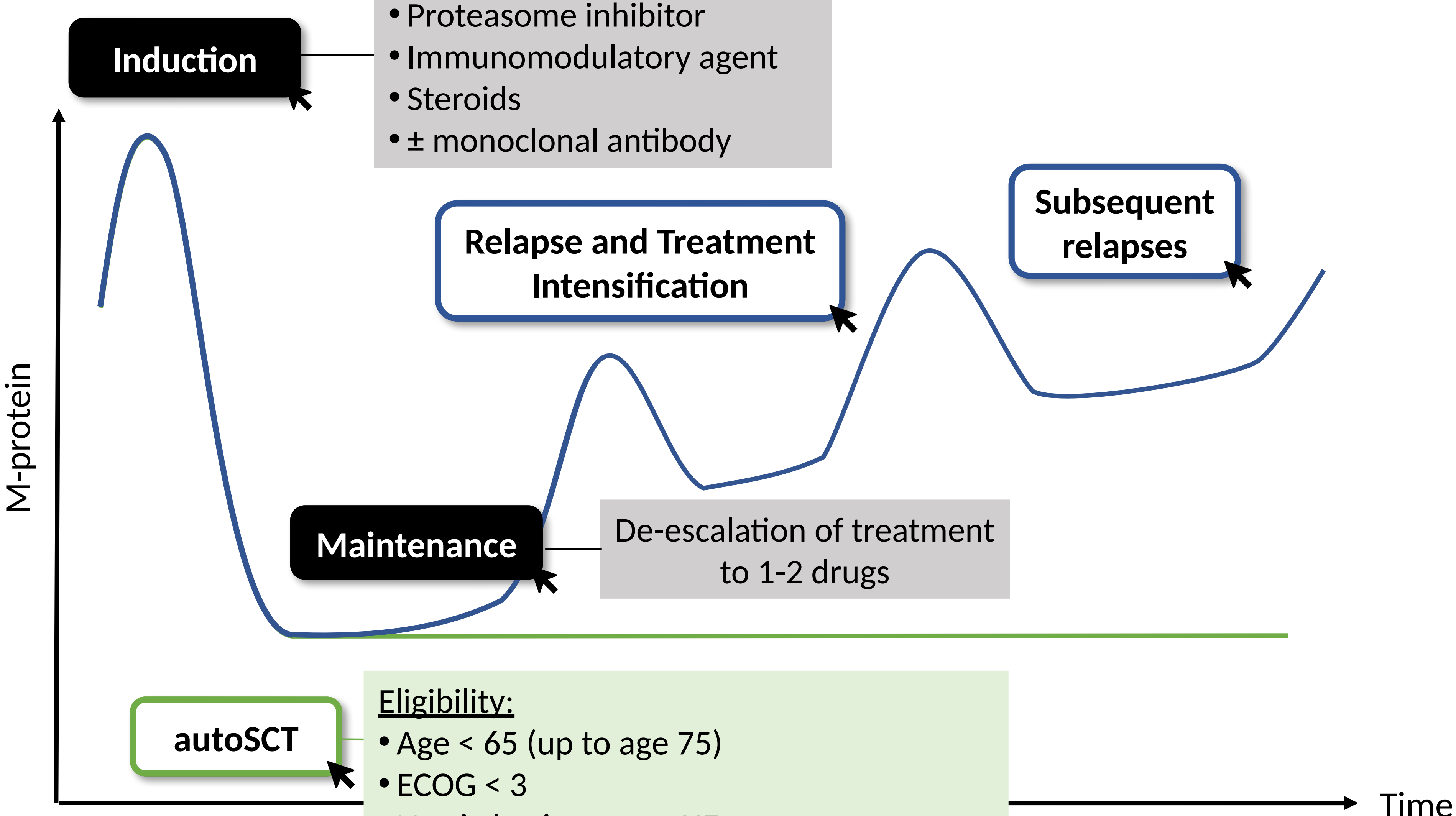
<u>Tumor factors</u>
Proliferation rate
Plasma cell leukemia
Extramedullary disease



<u>Patient factors</u>
Age
Comorbidities
Renal function
ECOG (performance status)

- Pathogenesis
- Testing
- Differential
- Prognosis
- **Treatment**
  - **Natural Hx**
  - Med Tox
  - Comp
- Cases

Natural History



Typically, 3 or 4 drug regimen:
- Proteasome inhibitor
- Immunomodulatory agent
- Steroids
- ± monoclonal antibody

De-escalation of treatment to 1-2 drugs

Eligibility:
- Age < 65 (up to age 75)
- ECOG < 3
- No cirrhosis, severe HF
- Renal disease is **NOT** a contraindication

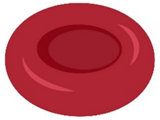
- Pathogenesis
- Testing
- Differential
- Prognosis
- Treatment
  - Natural Hx
  - Med Tox
  - Comp
- Cases

Complications of Therapies

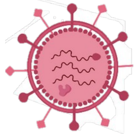
	Proteasome Inhibitors Bortezomib (Velcade) Carfilzomib (Kyprolis)	Immuno-modulatory Agents Lenalidomide (Revlimid) Pomalidomide (Pomalyst)	Steroids Dexamethasone	Monoclonal Abs Daratumumab Isatuximab Elotuzumab
Myelo-suppression	✓ ↓ PMNs/ plts	✓ ↓ PMNs/RBCs/plts	✗	✓
GI irritation	✓	✓	✓ Consider PPI	✓
Infections	✓ Shingles ppx	✓	✓ Consider PJP ppx	✓
Other Side Effects	**Neuropathy** Velcade **Cardiovascular** Carfilzomib	**VTE/arterial thrombosis** Asa vs. anticoag ppx	**Hyperglycemia** **Bone loss** Ca-Vit D or bisphosphonates	**Acute infusion reaction**

Pathogenesis
Testing
Differential
Prognosis
Treatment
Natural Hx
Med Tox
Comp
Cases

Treatment of Complications



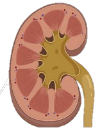
Anemia → Evaluate and treat for other causes



Immunodeficiency → IVIg if IgG is low (<500) + recurrent infections

↑ Ca^{2+} Hyper**C**alcemia → Bisphosphonates
Acute hospitalization

Boney lytic lesions → Ca and vit D (if Ca normal)
Systemic therapy + bisphosphonates
Consider radiation
Avoid surgery unless emergency



Renal disease → IVF to target high UOP
Consider biopsy
Urgent systemic therapy

Pathogenesis

Testing

Differential

Prognosis

Treatment

Cases

1 & 2a

2b & 2c

Case 1

55 year-old recently admitted for AKI and proteinuria presents to clinic for hospital follow-up. AKI resolved on discharge. SPEP and FLC ordered for proteinuria shows:

Test	Result (g/dL)	Reference (g/dL)
Albumin	4.2	3.6-5.4
Alpha1 globulin	0.3	0.1-0.3
Alpha2 globulin	0.6	0.5-0.3
Beta globulin	1	0.5-1
Gamma globulin	3 (H)	0.5-1.4

The monoclonal protein measures 2 g/dL

IFX: "IgG kappa monoclonal protein identified"

Kappa: 6.3 Lambda: 2.1 K/L Ratio: 3

How would you interpret these results?

There is an M-spike that measures <3 g/dL. FLC ratio is elevated.

What other tests would you order to aid in diagnosis?

CBC, calcium, creatinine, imaging, bone marrow biopsy.

...Labs: Hgb 13. Calcium 9. Cr stable at 1.
PET-CT shows no lytic lesions. Bone marrow biopsy shows 5% plasma clonal cells.

What is the diagnosis?

MGUS

Case 2a

64 year-old presents for follow-up of abnormal labs. SPEP, immunofixation, and K/L FLC assay results show:

Test	Result (g/dL)	Reference (g/dL)
Albumin	4.2	3.6-5.4
Alpha1 globulin	0.3	0.1-0.3
Alpha2 globulin	0.6	0.5-0.3
Beta globulin	1	0.5-1
Gamma globulin	4.5 (H)	0.5-1.4

The monoclonal protein measures 3.5 g/dL

IFX: "IgA lambda monoclonal protein"

Kappa: 1 Lambda: 5 K/L Ratio: .2

Labs: Hgb 14, Cr 1.0, calcium of 9.5

PET/CT: no lytic lesions

Bone marrow biopsy: 12% involvement.

What plasma cell dyscrasia does this patient have?

Smoldering MM. M-spike > 3 g/dL and > 10% bone marrow involvement but no end organ damage.

What is this patient's risk of progression every year?

10% for first 5 years and 5% per year for next 5 years

What lab monitoring should this patient have?

- Serial SPEP, IFX, UPEP, serum FLC assay, CBC, Ca, and Cr.
- Repeat at frequent intervals (~3 months initially), which may reduce over time if disease stable.

Pathogenesis

Testing

Differential

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Cases

1 & 2a

2b & 2c

Case 2b

One year later, the patient presents to the ED with acute onset back pain. Lumbar XR shows two vertebral fractures and several other boney lucencies. Labs shows Cr of 2.1 (baseline 1.1), Ca is 10.5.

What plasma cell dyscrasia does this patient have?

IgA lambda multiple myeloma. They now have signs of end-organ damage (boney lesions).

What studies would you order next?

Repeat SPEP, UPEP, serum FLC. Bone marrow biopsy. Albumin, beta-2 microglobulin, LDH are all helpful in staging and prognosis.

…A bone marrow biopsy, which shows no high-risk features. R-ISS stage is 1. They are started on RVd (lenalidomide, bortezomib and dexamethasone)

What other supportive/prophylactic medications does this patient need?

- Bisphosphonate
- Acyclovir ppx when on bortezomib
- ASA (or full anticoagulation) when on lenalidomide
- PJP ppx, PPI, vitamin D and calcium if continuing steroids

Case 2c

After treatment with RVd, they went into remission and was maintained on lenalidomide for 1 year. They represent with AKI with Cr up to 2.3. Subsequent testing revealed an increase in IgA lambda monoclonal spike. Kidney biopsy shows tubular deposition of light chain casts.

How would you treat the AKI?

This is **myeloma kidney**. Aggressive **IVF** (if no contraindication) to target ~3 L of UOP/day to prevent further cast deposition in tubules. Need to start urgent salvage systemic therapy

…They received second line therapy and subsequently underwent an autoSCT. They remained in remission for a few years until they represented to the ED with fevers, cough, hypoxia. A CT shows bilateral ground glass opacities.

What are the potential sources of this patient's presentation?

Common causes of CAP (*S.* pneumo, COVID-19, viruses), **Opportunistic pathogens** (PJP, aspergillus, endemic fungi) and **Nosocomial infections** (*Staph* and *Pseudomonas*). It is rare, but in extremely advanced myeloma, the infiltrates could also be **cancer**.

What additional therapy can we offer this patient if they present with repeat infections?

IVIg is indicated for patients with MM and multiple infections.

Newer Context — Updates Since 2023

Standing diagnostic framework (SPEP/IFX + serum FLC, CRABI, MGUS→SMM→MM thresholds) is unchanged. Updates since 2023 are in therapy and prognosis:

Front-line induction is now a QUADRUPLET: add an anti-CD38 mAb (daratumumab) to VRd. D-VRd reduced progression/death ~60% vs VRd — PERSEUS, NEJM 2024; FDA-approved for transplant-eligible NDMM (Jul 2024); Isa-VRd approved transplant-ineligible (IMROZ, 2024).

R2-ISS (D'Agostino, JCO 2022) refines R-ISS by adding gain/amp 1q21 (+ LDH, del17p/t(4;14)) into a 4-tier additive score — better separates the large R-ISS stage II group.

High-risk smoldering myeloma: daratumumab monotherapy delayed progression (5-yr PFS 63% vs 41%) — AQUILA, NEJM 2024; supports earlier intervention vs active monitoring in select high-risk SMM.

Relapsed/refractory: BCMA-targeted classes now standard — CAR-T cilta-cel (CARTITUDE-4, FDA Apr 2024, ≥1 prior line) & ide-cel; bispecifics teclistamab (Oct 2022), elranatamab, talquetamab (GPRC5D).

Monitoring: mass spectrometry (MALDI-TOF) emerging as a more sensitive alternative to immunofixation (not yet routine).

All updates clinician/heme-onc–reviewed before teaching. Sources: PERSEUS NEJM 2024; D'Agostino JCO 2022; AQUILA NEJM 2024; FDA approvals 2022–2024.

Take-Home Points

MGUS, smoldering myeloma (SMM), and MM are a spectrum of clonal plasma-cell disorders; complications follow CRABI — hyperCalcemia, Renal dysfunction, Anemia, Boney lytic lesions, Immunodeficiency.

Screen with SPEP + serum immunofixation AND serum free light chains (FLC). FLC catches the ~16% of light-chain-only myeloma SPEP misses; SPEP+FLC sensitivity ~99% and replaces the 24-h UPEP.

Distinguish MGUS / SMM / MM by M-protein size, % clonal marrow plasma cells, and end-organ damage (CRABI) or a myeloma-defining biomarker (≥60% marrow, FLC ratio ≥100, >1 focal MRI lesion).

Progression risk: MGUS ~1%/yr; SMM ~10%/yr for the first 5 years. Order β2-microglobulin (mg/L), LDH, albumin, and FISH at diagnosis for R-ISS / R2-ISS staging.

Front-line therapy is now quadruplet (anti-CD38 mAb + VRd) ± autoSCT then maintenance; manage hypercalcemia, cast nephropathy (aggressive IVF, urgent systemic therapy), and infection/VTE prophylaxis.

Attribution & Optimization Changelog

Original lesson: “Plasma Cell Dyscrasias,” TeachIM.org (Published Jan 2023).

Authors: Meryl Colton MD MSc; Peter Forsberg MD. Section Editor: Christopher Geiger MD. Executive Editor: Yilin Zhang MD. University of Colorado.

https://teachim.org/teaching_material/plasma-cell-dyscrasias/

Optimization (2026-06-10) — surgical corrections, animations/graphics preserved:

- Slide 8 (R-ISS): β2-microglobulin units “g/dL” → “mg/L” (×2; β2M is reported in mg/L — lesson page uses mg/L); albumin g/dL left unchanged.
- Slide 8: R-ISS stage III median survival “42 months” → “43 months” (matches lesson page table & Palumbo, JCO 2015).
- Slides 9–10 (incl. notes): spelling “Proteosome” → “Proteasome”.
- Appended: this attribution slide, a take-home slide, and a 2023–2026 newer-context slide.

Human review: Slide 4 & case tables list Alpha-2 globulin reference range as “0.5-0.3” (reversed/implausible); intended value not stated, left unedited — confirm correct range. Medical content should be clinician-reviewed before teaching.